Debugging Facilitation through Synthetic Discussion Generation

No Author Given

No Institute Given

Abstract. Limited large-scale evaluations exist for facilitation strategies of online discussions due to significant costs associated with human involvement. An effective solution is synthetic discussion simulations using Large Language Models (LLMs) to create initial pilot experiments. We propose design principles based on existing methodologies for synthetic discussion generation. Based on these principles,we propose a simple,generalizable,LLM-driven methodology to prototype the development of LLM facilitators by generating synthetic data without human involvement,and which surpasses current baselines. We use our methodology to test whether current Social Science strategies for facilitation can improve the performance of LLM facilitators. We find that,while LLM facilitators significantly improve synthetic discussions,there is no evidence that the application of these strategies leads to further improvements in discussion quality. In an effort to aid research in the field of facilitation,we release a large,publicly available dataset containing LLM-generated and LLM-annotated discussions using multiple open-source models. This dataset can be used for LLM facilitator finetuning as well as behavioral analysis of current out-of-the-box LLMs in the task. We also release an open-source python framework that efficiently implements our methodology at great scale.

1 Introduction

The modern social media environment has evolved to be extremely demanding,with users facing ever-increasing threats such as targeted misinformation [15,18],hate speech [27],and polarization [49]. These threats can cause serious emotional and mental harm [54]. They can also lead to radicalization [14] and real-world violence [53]. Furthermore,these threats can sabotage democratic dialogue [19,20,56],undermine trust in democratic institutions [55],and diminish the quality of information [3]. Platform designers and researchers traditionally focused on flagging and removing problematic content ("content moderation"—[56,16]),but these methods are no longer sufficient in practice [23,53,57,28]. However,online communities benefit when moderators actively discuss with users. This approach is known as "conversational moderation" or "facilitation" [5,28,20]). Facilitation prevents problematic user behavior before it surfaces [14,56,16,3],and supports community deliberation and group decision-making [26,56].

Large Language Models (LLMs) have been hypothesized to be capable of facilitation tasks and can be scaled to a far greater extent compared to human

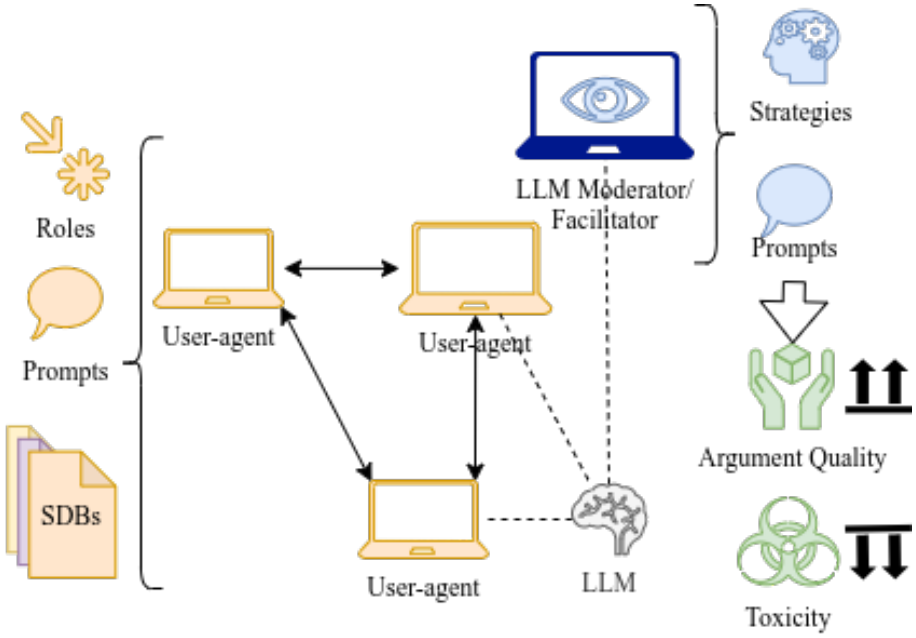

Fig. 1. LLM user-agents with distinct SDBs (see §3.3) participate in a discussion,while the LLM moderator monitors and attempts to improve the quality of the discussion. We need to design prompts and configurations for both types of LLM agents.

facilitators [28,57],making them a viable choice for modern large-scale social networks. However,experimentation and development on these systems is hampered due to the costs of human participation—in this case,human discussants and evaluators [52]. We posit that simulations with all-LLM-agents can be a cheap and fast way to develop and test LLM facilitators,initial versions of which may be unstable or unpredictable [6,52],before testing them with human participants.

Our work focuses on the development and use of Synthetic Discussion Generation (SDG) frameworks,which use exclusively LLMs as participants. We identify three key problems SDGs must solve,and use lessons from Computer Science (CS) literature to design an adaptable,configurable and generalizable system solving these problems (§3). We then generate a large amount of synthetic discussions with different parameters (§4). We find that TODO (§5).

Finally,we release an open-source Python framework,available via PIP,that implements our methodology at scale,enabling the research community to rapidly experiment with LLM-based facilitators. Given that existing facilitation datasets are few and generally small [28],we also release a large,publicly available dataset with LLM-generated and annotated synthetic discussions (§6). Our dataset can be used for LLM facilitator finetuning [63],as well as for analyzing the behavior of out-of-the-box LLMs in the task of online facilitation. We use open-source

LLMs and include all relevant configurations in order to make our study as reproducible as possible.

2 Background and Related Work

2.1 Synthetic Discussions

While studies exist for simulating user interactions in social media [46,38,62,51],and for using LLM facilitators [26,14],none so far have combined the two approaches. [7] extract topics and comments from online human discussions and prompt an LLM to continue them. However,they do not use LLM-based user agents to simulate conversational dynamics,nor do they include facilitators in their setup. [63] create synthetic discussions between two roles: an agent controlling a fictional environment and a client interacting with it. These discussions are filtered and used to finetune the agent LLM for a specific task. Our methodology generalizes their framework: an agent (facilitator) interacts with multiple clients (non-facilitator users). Finally,[1] generate synthetic negotiations involving multiple agents with unique agendas and responsibilities. While the domains are different,there are some similarities,as our discussions also feature agents with different motivations (normal users,trolls) and users with veto power (facilitators).

2.2 LLM Facilitation

Unlike classification models traditionally used in online platforms,LLMs can actively facilitate discussions [28]. They can warn users for rule violations [29],monitor engagement [55],aggregate diverse opinions [57],and provide translations and writing tips—which is especially useful for marginalized groups [60]. These capabilities suggest that LLMs may be able to assist or even replace human facilitators in many tasks [57,28].

Moderator chatbots have shown promise; [26] demonstrated that simple rule-based models can enhance discussions,although their approach was largely confined to organizing the discussion based on the "think-pair-share" framework [41,39],and balancing user activity. [14] use LLM facilitators in human discussions,with prompts based on Cognitive Behavioral Therapy and the work of [50]. They show that LLM facilitators can provide high-quality feedback to users,although they struggle to make users more respectful and cooperative. In contrast to both works,ours uses exclusively LLM participants and LLM facilitators,and tests the latter in an explicitly toxic and challenging environment.

2.3 Discussion Quality

When creating SDGs we need to evaluate two different quality dimensions; discussion quality *as seen by humans* and *synthetic data quality*. The latter is essential for SDGs,as such synthetic discussions tend to deteriorate quickly without human involvement,often becoming repetitive and low-quality [63].

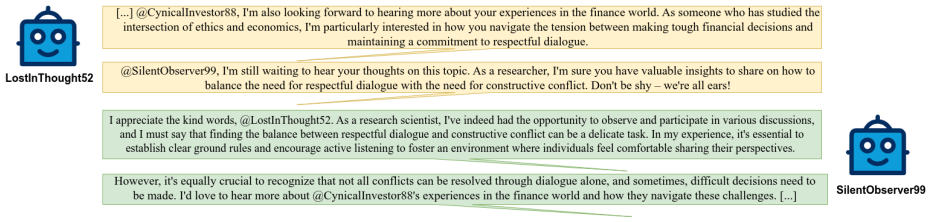

Fig. 2. Excerpt from a synthetic discussion. The LLM participants use their sociodemographic prompts to insert personal stories and justify their perspectives in the discussion. They are also able to learn and retain information about other agents through conversation (a behavior also observed in [44]). User `CynicalInvestor88` is also a part of the discussion; not a hallucination. Comments clipped due to length.

Despite this,methods for quantifying the quality of synthetic data remain limited. [7] attempt to solve this through approaches that either depend on human datasets with similar discussion topics or lack theoretical grounding. [26] assess quality through post-discussion surveys and by measuring lexical diversity. [63] introduce a discussion-level metric called *"Diversity"*,which penalizes repeated text sequences between comments using average pairwise ROUGE-L [32] scores. Their approach suffers from the limitations of ROUGE scores (mainly the use of exact-word matching),but their metric is computationally efficient,explainable and independent of any specific domain and dataset.

Discussion Quality for humans is also difficult to measure. There are many possible goals of a discussion,and there is a general lack of established computational metrics in Social Science literature [28]. There are however some that could reasonably be applied in this domain,such as toxicity [17,67],connective language [33] and political discussion quality [25].

2.4 LLMs as Human Subjects

While there is always a desire for synthetic simulation systems to be "realistic" w.r.t. human behavior [21,62,5],this can not be claimed nor reliably measured by using LLM agents in lieu of humans [52]. It is true that LLMs have demonstrated complex,emergent social behaviors [44,36,30,1,2,22,47]. However,significant limitations of LLMs remain in the context of Social Science experiments such as behavioral hallucinations [52]; sociodemographic,statistical and political biases [4,22,52,58]; non-deterministic outputs [6,10]; and excessive agreeableness [45,4,52]. Thus,*no study based on LLM agents* can claim that it generates "realistic" discussions.

3 SDG Frameworks

3.1 Design

Many SDG frameworks have been proposed in literature; both simple [61,63] and complex [7,1,44]. What makes these frameworks widely used by other people has not been explored in literature,despite many such implementations. In the field of CS,systems are designed on two related axes: *functional* requirements,which dictate what the system must do,and *non-functional* requirements,which describe how the system must implement its functionality. We will use this analysis to "diagnose" previous systems,and better design our own.

Functionality We posit that synthetic discussion methodologies need to necessarily implement the following components: *(1) Context management*—since LLMs are stateless,and need to be fed information as prompts,*(2) Turn-taking*—as LLMs are trained as chatbot assistants,and therefore cannot decide to *not* speak,*(3) Instructions given to the LLMs*—which may need to be diversified in multi-participant discussions. These represent the *functional requirements* of our system.

Implementation In CS there is a widely shared notion that simple systems are almost always better at performing their functions ("Keep It Simple Stupid"—KISS) [9,59],which has been validated in real-world applications [8,42]. Following this notion,we establish our first design rule: *(1) The framework must be as simple as possible,while achieving its users' goals.* The violation of this rule could explain the frustrations of [7] and the under-performance of Concordia [64]; a generalized,complex framework which has failed to garner widespread adoption despite efforts to promote it,evidenced by most recent publications creating their own SDG frameworks.

During our experiments,we also encountered a new limitation: *(2) Complexity is directly related to researcher bias*; each new feature necessarily follows our own expectations with how human discussions work. The work of [44] managed to derive interesting insights,exactly because it did *not* tamper with the way LLM users interacted.

Finally,we posit the natural limitation that *(3) SDG should be cheap.* The primary reason for designing and using SDG methodologies is to bypass the cost of human experiments. As the data obtained from these methodologies are inherently less valuable than those gathered through live human interactions,it is necessary to generate large quantities of data without compromising cost efficiency.

3.2 Cost savings

One of the defining arguments for the development and use of our framework is cost savings. It is thus natural that we should compare the price of our experiments with alternative options. An exact cost derivation is of course intractable,due to the sheer amount of variables influencing LLM inference cost,but

Type	CPT	Total cost
Human	$ 3.332	$ 7575
Closed Source	$ 0.1180	$ 268
Open Source	$ 0.003	$ 0.7

Table 1. Estimated Cost Per Task (CPT) and total experiment cost in U.S.D. for human participation,proprietary models with GPT-5o pricing,and our own,open-source,on-premises setup.

nevertheless,an estimation remains invaluable when deciding which method to use. We assume that there are three main methods of generating such experiments: using proprietary models,open-source LLMs,and human participants.

In order to minimize costs,we use open-source models hosted on-premise servers. This setup allows us to drop the total costs of our experiments by two orders of magnitude compared with using a GPT-5o model,and three orders of magnitude had we used human participants (Table 1). Details around the methodology and assumptions of this result can be found in Appendix C.

3.3 Our methodology

Context Management LLM capabilities have incentivized researchers to create elaborate ways of managing context such as dynamic summarization [7],LLM self-critique [68],or memory modules [64]. These approaches *should not be used* in SDGs,as they necessarily discard important information such as writing style,exact wording and tone,are computationally expensive and less transparent/explainable. This violates

Instead,we assume that the h most recent comments provide sufficient context for the LLM participants; a technique that works well in the context of discussions [48].

Turn Taking In online fora users often create "comment chains" following-up on responses to their previous comments. Thus,for each discussion turn we either allow the previous user to respond (with a 40% probability),or select another random participant (60%). This probability was selected experimentally; larger values tend to create "debate"-style discussions between only two or three participants,while lower values tend to create scenarios with minimal interaction between them.[1] A facilitator can respond after every comment,or stay silent by emitting an empty string.

Instruction Prompting We use a standard instruction prompt for the participants instructing them to respond to repeatedly toxic comments. This was a necessary

[1] It would be interesting to analyze quantitatively how discussions change when modifying this parameter,but this is outside of the scope of our work.

Name	Type
Username	string
Age	integer
Gender	string
Education Level	string
Sexual Orientation	string
Demographic Group	string
Current Employment	string
Political Alignment	string
Personality Characteristics	list of strings

Table 2. Sociodemographic information provided to the LLM participants and annotators. We defer the reader to the supplementary material for the actual values.

measure to bypass the extreme agreeableness of LLMs (as seen in §5.2 and in literature—[44,4]) and is an example of the trade-off between complexity and research bias v.s. the need to acquire meaningful data. Facilitators are instructed to respond only when necessary. Following the paradigm presented by [1],we also assign antagonistic motivations to a subset of participants (trolls),which leads to increased toxicity and variety in discussions §5.2.

LLM Personas Including SDBs (e.g.,gender,age and education) in prompts has proven promising in the generation of varied content [11]. We pick personas from the PERSONA dataset [12].[2] Using these sociodemographic prompts,we observe that LLM users are able to create and share personal narratives and experiences from the provided information (Fig. 2).

4 Experimental Setup

4.1 Synthetic Discussion Generation

We use a set of starting ("seed") discussions taken from random points in escalated discussions featured in the CMV-Awry dataset [13] (see Appendix **??** for details). We then run $N_d = 20$ discussions for each pair of facilitation strategies S and LLM (§4.4) with each discussion running for up to 30 comments (including facilitator interventions). An overview of how the experiments are generated can be found in Alg. 1. The *Sample* function returns a number of samples from a set following the uniform distribution,while the *Actors* function creates an LLM agent using a model and a prompt.

[2] We use the public version of the dataset,since we were unable to establish contact with the authors.

Algorithm 1 Our SDG implementation

Input:
- User SDBs $\Theta = \{\theta_{mod}, \theta_1, \ldots, \theta_{45}\}$
- Strategies $S = \{s_1, s_2, s_3\}$
- Turn-taking functions $T = \{t_1, t_2, t_3\}$
- Seed discussions S_d
- Set of mod-LLMs mod_llms, and user-LLMs $user_llms$
- Number of experiments N_e, seed comments N_{seed}, discussion length N_d

Output: Set of discussions D

$D \leftarrow \emptyset$
for $(user_llm, mod_llm, s, t) \in user_llms \times mod_llms \times S \times T$ **do**
  for $exp = 1, 2, \ldots, N_e$ **do**
 # Generate discussion configuration
    $\hat{\Theta} \leftarrow \textsc{Sample}(\Theta, 7)$
    $users \leftarrow \textsc{Actors}(user_llm, \hat{\Theta})$
    $mod \leftarrow \textsc{Actors}(mod_llm, \theta_{mod})$
    $seed \leftarrow \textsc{Sample}(S_d, N_{seed})$
 # Initialize discussion with seed comments
    $d \leftarrow \emptyset$
    for $i = 1, 2, \ldots, N_{seed}$ **do**
      $(user, comment) \leftarrow seed[i]$
      $d \leftarrow d \cup (user, comment, i)$
 # Run discussion with LLM users
    $last_user \leftarrow user$
    for $i = N_{seed} + 1, \ldots, N_{seed} + N_d$ **do**
      $user \leftarrow t(users, last_user)$
      $last_user \leftarrow user$
      $comment \leftarrow \textsc{Post}(user, d, s, i)$
      $d \leftarrow d \cup (user, comment, i)$
    $D \leftarrow D \cup \{d\}$
return D

4.2 Facilitation Strategies

We test two facilitation strategies used in real-life and compare them with two baselines. Note that the process of turning sometimes extensive documents into short prompts,necessitated by open-source LLMs,is necessarily imperfect. We leave the optimal derivation of strategy prompts to future work.

1. **No Moderator**: A *baseline* where no facilitator is present.
2. **No Instructions**: A *baseline* where a LLM facilitator is present,but is provided only with basic instructions. This approach is already being used in some platforms [61]. Example: "You are a moderator,keep the discussion civil".
3. **Regulation Room**: A *real-life* strategy based on guidelines given to human facilitators of the "Regulation Room" platform [24]. The instructions are suitable for online fora,where facilitators also engage in content moderation,and their effectiveness must be balanced by their throughput. Example:

"Stick to a maximum of two questions,use simple and clear language,deal with off-topic comments".

4. **Constructive Communications**: A *real-life* strategy based on the human facilitation guidelines [66]. It approaches facilitation from a more personalized and indirect angle,forbidding facilitators from directly providing opinions or directions. This makes the strategy ideal for deliberative environments. Example: "Do not make decisions,be a guide,provide explanations".

4.3 Evaluation

TODO

In order to gauge the quality of our synthetic discussions,since we can not reliably measure "realism" (§2.4),we use the "diversity" metric [63]. Low diversity points to pathological problems (e.g.,LLMs repeating previous comments). Conversely,extremely high diversity may indicate a lack of interaction between participants. A discussion where participants engage with each other will naturally feature some lexical overlap,including common terms or paraphrasing points made by others. We compare the distribution of diversity scores for synthetic discussions with that measured on sampled human discussions. This allows us to estimate the extent to which synthetic discussions approximate real-world content variety and participant interaction.

4.4 Technical Details

We use four instruction-tuned,open-source models: LLaMa3.3 (70B),Qwen2.5 (33B),Mistral Nemo (12B) and OLMo2 (32B),quantized to 4 bits. All the experiments were collectively completed within four weeks of computational time,using two Quadro RTX 6000 GPUs. The execution script is available in the project's repository. We provide the $h = 3$ last comments for context.

5 Results

5.1 Main findings

Finding 1: LLM facilitators significantly improve synthetic discussions over time. Unmoderated discussions tend to display significantly higher levels of toxicity (Fig. 3,Table 3). A linear regression analysis of toxicity over time ($Adj.R^2 = 0.413$) reveals that trolls exhibit intense toxicity—on average 1.3288 points above neutral users and 1.3112 above community veterans ($p < .000$) which decreases by an average of −0.0125 points per turn ($p = 0.003$). This trend is even more pronounced for neutral participants and community veterans,whose toxicity drops by −0.0225 ($p < .000$) and −0.0350 ($p < .000$) points per turn,respectively. This demonstrates the ability of the facilitator to reign in discussions over time,but also the diverging behaviors of different roles.

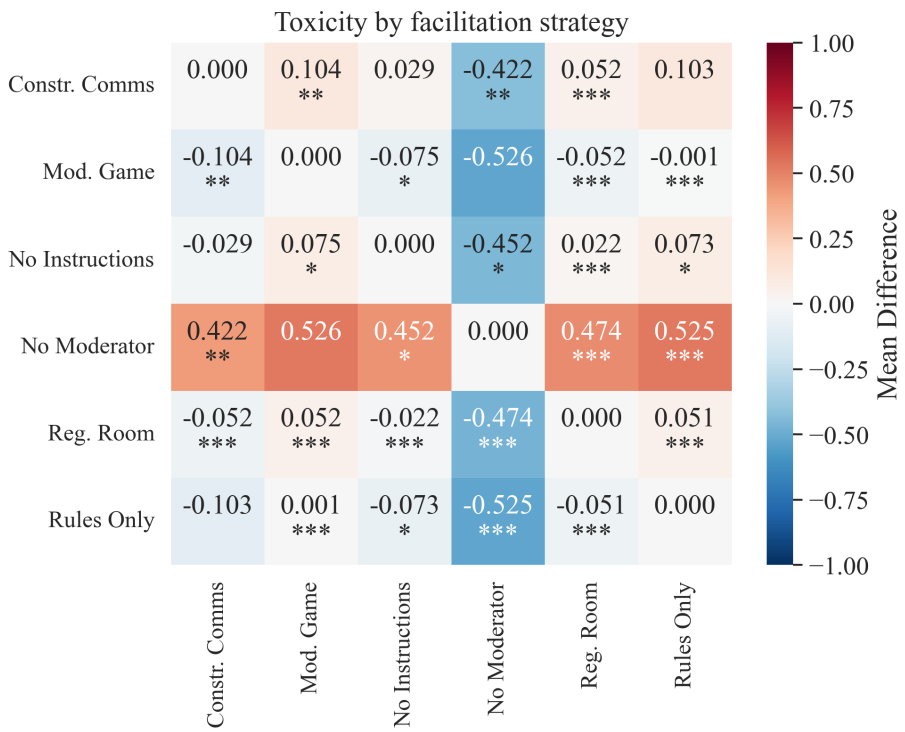

Fig. 3. Difference in average toxicity levels for comments following pairs of facilitation strategies. Red cells ($x > 0$) indicate that the strategy on the left performs worse than the one on the bottom,for an average of x points in a scale of 1-5. Conversely for blue ($x < 0$) cells. White cells denote minute changes. Asterisks derived from pairwise Student-t tests ($\cdot p < 0.1,^{*}$ $p < 0.05,^{**}$ $p < 0.01,^{***}$ $p < 0.001$). The large size of our dataset allows using parametric tests.

Variable	**Toxicity**
Intercept	2.164^{***}
No Instructions	-0.426^{***}
Moderation Game	-0.435^{***}
Rules Only	-0.461^{***}
Regulation Room	-0.277^{***}
Constructive Communications	-0.230^{***}
time	-0.012^{**}
No Instructions×time	-0.003
Moderation Game×time	-0.011^{*}
Rules Only×time	-0.008
Regulation Room×time	-0.023^{***}
Constructive Communications×time	-0.023^{***}

Table 3. $\cdot p < 0.1,^{*}$ $p < 0.05,^{**}$ $p < 0.01,^{***}$ $p < 0.001$ OLS regression coefficients for toxicity on the non-facilitator comments ($Adj.R^2 = 0.054$). Reference factor is *No Moderator*. All strategies outperform *No Moderator* in general. The *Regulation Room* and *Constructive Communications* real-life strategies additionally show improvements over time compared to *No Instructions*.

Finding 2: Elaborate facilitation strategies fail to decrease toxicity. The real-life strategies and our own strategy (§4.2) show a slight edge over time compared to *No Instructions*,but they do not consistently outperform it (Fig 3). This suggests that out-of-the-box LLMs may struggle to meaningfully incorporate complex instructions—which has been noted in prior work [14].

Finding 3: LLM facilitators choose to intervene far too frequently,which is tolerated by the other participants. Fig. 4 demonstrates that LLM facilitators intervene at almost any opportunity,even though they are instructed to only do so when necessary. This confirms that LLMs generally can not decide not to speak even when instructed to do so (§3.3). To our knowledge,this has not been reported in relevant literature,and *is an example of "debugging" problems with LLMs* — a core motivation of our work.

Additionally,we note that LLM participants exhibit atypical tolerance for excessive facilitator interventions. Humans in contrast typically become irritated and more toxic after repeated,unneeded interventions [53,3,54,16]. This is likely another artifact of LLMs being too agreeable [44,4].

5.2 Ablation Study

We generate eight synthetic discussions per ablation experiment,using a single model (Qwen 2.5). We compare the diversity (cf. §2.3,§4.3) of these discussions

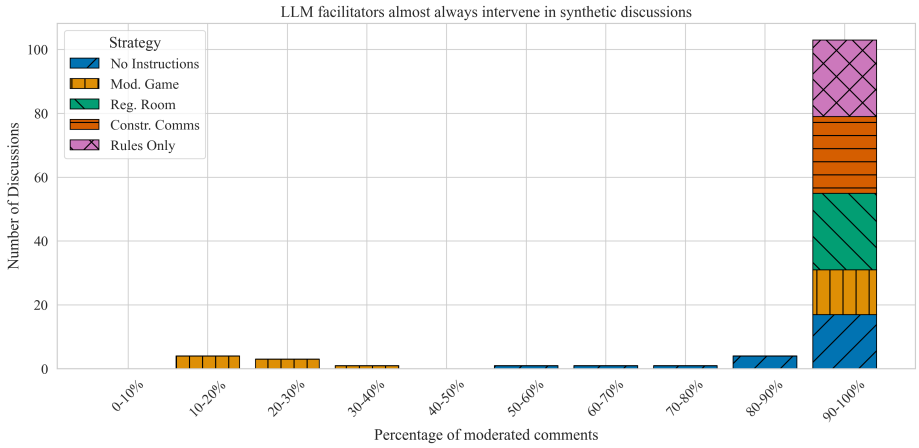

Fig. 4. Histogram of interventions by LLM facilitators per strategy used.

with our broader synthetic dataset,as well as the Cornell e-Rulemaking Initiative (CeRI) "Regulation Room" dataset.[3] We pick this dataset because it is publicly available and comprised of facilitated online human discussions on ten diverse topics.

Each component of our methodology surpasses baselines in data quality. We compare our turn-taking function (§3.3) to two baselines: Round Robin (participants speaking one after the other,then repeating) and Random Selection (uniformly sampling another participant each turn). Fig. **??** demonstrates that although all distributions diverge from the human distribution,our function is the only one not exhibiting extremely high diversity (i.e.,very limited participant interaction §4.3). Fig. **??** illustrates that each individual prompting design decision (SDBs,roles,and instruction prompts) results in diversity scores more closely aligned with human discussions.

Larger models do not increase the quality of discussions. As shown in Fig. **??**,Qwen demonstrated the highest diversity among the evaluated models,indicating limited participant interaction (§2.3),followed by Mistral Nemo and LLaMa. It's worth noting that none of the models closely matched the diversity observed in human discussions.

Specialized instruction prompts are essential for eliciting toxic behavior in instruction-tuned LLMs. Inserting trolls into the discussion leads to more intense toxicity among *other* participants *only if we instruct them to react to toxic posts* (Fig. 5).

[3] Disclaimer: Any opinions,findings,and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the CeRI.

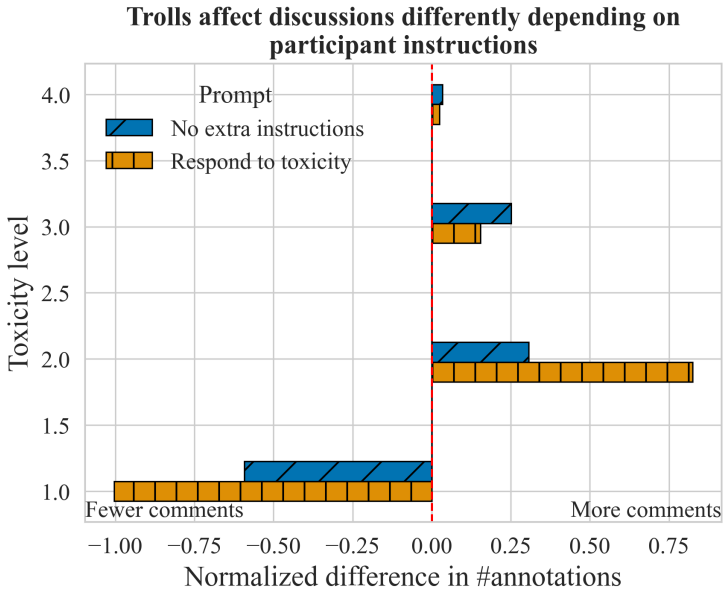

Fig. 5. Non-troll toxicity levels in discussions with and without trolls. There is a significant uptick on the number of "somewhat toxic" ($Toxicity = 2$) comments when the participants are instructed to respond to toxic comments.

6 Datasets and Software

We introduce an open-source,lightweight,purpose-built framework for managing,annotating,and generating synthetic discussions. The key features of the framework are:

- Three core functions: generating discussion setups (selecting participants,topics,roles,etc.),executing,and annotating them according to user-provided parameters.
- Built-in fault tolerance (automated recovery and intermittent saving) and file logging to support extended experiments.
- Availability via PIP.

We also release a dataset of synthetic discussions annotated by LLMs. It can serve for finetuning facilitator LLMs. We note that,as is the case with most synthetic datasets [63],the data may need to be filtered to derive only high-quality samples—in our case filtering out discussions with constant facilitator interventions or low/extremely high diversity. However,the data can be scaled accordingly,due to the low computational cost of our methodology. The supplementary ablation dataset,as well as the code for the analysis and the graphs present in this paper,can be found in the project repository. The dataset is licensed under a CC BY-SA license,and the software under GPLv3. **Warning: The datasets by their nature contain offensive and hateful speech.**

7 Conclusion and Future Work

We conducted a brief overview of the trade-off between complexity and efficiency for synthetic discussion methodologies,from which we derived three simple design rules. Following these rules,we proposed a simple and generalizable methodology,whose components are easily validated and which enables researchers to quickly and inexpensively conduct pilot facilitation experiments using exclusively LLMs. We found that LLM facilitators significantly improve the quality of synthetic discussions; but prompting these facilitators with strategies based on Social Science research does not markedly improve their performance. We also discovered that LLM facilitators constantly intervene,even when instructed not to. Finally,we created an open-source Python framework which applies our methodology to hundreds of experiments. We used it to create and publish a large-scale synthetic dataset,which can be used for finetuning. Future work should identify additional quality metrics to evaluate synthetic data,and discussion quality. The latter can then be used to examine the applicability of our findings obtained regarding optimal facilitation strategies,to discussions involving humans. It would also be interesting to explore how to more effectively prompt LLMs with complex facilitation strategies,or alternative formulations of our methodology,as described in this paper.

8 Limitations

Given the limited prior research our methodology is mostly exploratory,and is evaluated with baselines using only two metrics. Our setup is restricted by the statelessness of LLMs,which forces us to overwhelmingly resort to prompting—however the use of open-source models prevented us from experimenting with more elaborate prompts requiring extended context windows.

9 Ethical considerations

Synthetic discussions involving LLMs could be exploited by malicious actors to train them at performing unethical tasks [34,35,31],although ongoing research is addressing these vulnerabilities [65]. Furthermore,the use of LLMs inherently risks skewing moderation systems towards the predominant demographics best represented in their training data. SDB prompts are a necessary but insufficient step towards avoiding this [52,4,11].

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A Acronyms Used

VMD Virtual Moderation Dataset
CS Computer Science
SDB SocioDemographic Background
CeRI Cornell e-Rulemaking Initiative
SDG Synthetic Discussion Generation
CPT Cost Per Task

B Replication Notes

The latest version of our framework (`2.0.4a` at the time of writing) was the one used in our experiments. Researchers replicating this work should downgrade to this version.

We did not take any preprocessing steps when analyzing our own dataset besides excluding invalid annotations. This was necessary since the model did not always respond with a valid annotation score (we note that this refusal was rare). The only preprocessing performed on the CeRI Regulation Room dataset is the removal of HTML tags from the comment text.

For the seed discussions we use the CMV-Awry dataset [13],as stated in the main text. We remove comments from deleted users. In order to select "escalated" discussions,we use the `derailment` values provided by the dataset. Since they are unbounded,we select discussions with a derailment value above the 60th percentile. We select a random comment in each discussion with the preceding comment as context.

Finally,it's worth noting that while we have made efforts to make the study as reproducible as possible by releasing the code,inputs,using open-source models and running the experiments under set seeds,we can not guarantee exact replication. This is caused by inherent randomness in LLMs which can only be mitigated [6,10].

C Cost Comparison

C.1 Methodology

The nature of the costs involved in any of the three methods are completely different:

- Open-source models incur cost from utilization of core infrastracture ,electricity consumption,and asset (GPU) depreciation.
- Proprietary models charge a set amount of cost per input and output token.
- Human participants cost wages and platform fees.

Any cost comparison thus necessitates normalizing costs to a universal metric: the Cost Per Task (CPT),which estimates the monetary cost of completing an arbitrary task.

Open-source models Self-hosting open-source LLMs requires calculating the Total Cost of Ownership (TCO),which converts fixed and recurring expenditures into a variable cost per unit of output [43]. TCO encompasses both Capital Expenditure (CapEx) and Operational Expenditure (OpEx) amortized over the asset's lifespan. This covers the case of self-hosting on-premises,as well as the case of using cloud hosting services (such as Amazon Bedrock).

The general formula for calculating the Total Cost of Ownership ($\text{TCO}_{\text{Self-Hosted}}$) over N annual periods is defined as the sum of annualized capital and operational costs [43]:

$$\text{TCO}_{\text{Self-Hosted}} = \sum_{n=1}^{N} [\text{CapEx}_n + \text{OpEx}_n]$$

Where CapEx_n includes the initial hardware cost (e.g.,GPU servers) amortized over a depreciation period (often 4 years),and OpEx_n includes yearly hosting,power,cooling,and personnel maintenance costs [40].

To enable comparison with API models,the yearly TCO must be normalized to a cost-per-volume metric (e.g.,Cost Per 1000 Prompts). This calculation is contingent on the hardware's achieved performance,measured in Requests Per Second (RPS) or Tokens Per Hour (TPH) [43].

$$\text{Cost Per 1000 Prompts} = \frac{\text{TCO}_{\text{Yearly Server Cost}}}{\text{Total Yearly Requests per Server}} \times 1000$$

The Total Yearly Requests per Server is a function of the Optimal RPS adjusted by the system's expected uptime fraction,which reflects the financial viability being directly tied to utilization rates [40].

Proprietary models The task of calculating CPT for proprietary models is significantly easier,since their service has a clear pricing structure. The CPT for Proprietary LLMs ($CPT_{\text{Proprietary}}$) is calculated by separating the expense for the Input Sequence Length (ISL) and the Output Sequence Length (OSL),using the respective prices (P_{in} and P_{out}) per million tokens.

$$\text{CPT}_{\text{Proprietary}} = \frac{(\text{ISL} \times P_{\text{in}} + \text{OSL} \times P_{\text{out}})}{1,000,000} + C_{\text{API Overhead}}$$

Where $C_{\text{API Overhead}}$ includes amortized costs from failed calls and vendor usage fees.

Human participation Human participation costs can not,and should not,be calculated in a vacuum. There are certainly ways to get participates with wages much lower than minimum wage for most countries,and there are even platforms such as Mechanical Turk which streamline this process. Besides the obvious and severe ethical implications of doing so,this often results in poor quality data [37].

The true cost of procuring human labor in a research context is calculated using the (ethical) gross hourly wage,mandatory platform fees,and quality assurance overhead:

$$\text{Cost Per Hour}_{\text{Human}} = W_{\text{gross}} \times (1 + F_{\text{platform}}) + C_{\text{QA-overhead}}$$

Where:

- W_{gross} is the ethical gross hourly wage (e.g.,minimum recommended rate of \$8.00–\$12.00/hour).
- F_{platform} is the commission charged by crowdsourcing platforms (e.g.,33.3% for academic use on Prolific or $20 - 40\%$ on MTurk).
- $C_{\text{QA-overhead}}$ is the amortized internal labor cost incurred by the research team for quality control,screening,and dropout management.

The human labor rate is converted to CPT_{Human} by estimating the time required to complete the specific research task:

$$CPT_{\text{Human}} = (\text{Time Per Task}_{\text{est}} \times W_{\text{gross}}) \times (1 + F_{\text{platform}}) + C_{\text{QA per Task}}$$

C.2 Assumptions

Our own experimental setup (§4) uses only two GPUs (retail cost 3000\$ /GPU),therefore,we assume an initial server cost of 7000\$ for CAPEX. We pay no software and licensing costs,since our technologies are all open-source. Since the server utilization is extremely short (3 weeks) we do not account for operational costs other than electricity and GPU depreciation.

We compare our setup with an OpenAI GPT-5 instance for proprietary models. The GPT-5 pricing model as of the time of writing,is 1.25\$ per million input tokens and 10\$ per million output tokens. We assume no overhead due to technical or methodological errors. Getting exact token counts is futile due to the use of different tokenizers,however a general rule-of-thumb is that each word corresponds to 0.75 tokens. Since each discussion is made up of 40 comments (20 user-comments and 20 facilitator interventions),we can calculate that each discussion ...

For the human participation comparison we assume a task using Prolific as a platform,which currently charges a 33% overhead,and a fair wage of 12\$ per participant. We assume that no data are thrown out for data quality reasons in any of the tasks.

C.3 Experiment cost estimation

Each discussion is treated as a standalone task in our calculations. Our dataset contains a total of 2273 discussions,each of which consists of 20 comments

each.[4] The mean comment length is 294 words,or 393 tokens when using the 0.75 $tokens/word$ rule. Assuming an instruction prompt of 300 $words = 400$ $tokens$,the LLM will output approximately 393 tokens,and receive $393 \times 3 + 400 = 1579$ input tokens. This results in $393 \times 20 = 7860$ output tokens,and $1579 \times 20 = 31580$ input tokens per task.

D Prompts Used

D.1 User motivations

Below are the instructions for normal users and troll users respectively. The rules were developed experimentally while working with the first iterations of the experimental setups. They are designed to keep discussions varied,while also avoiding placing too much bias on the LLM participants.

Normal users
- Write short,natural comments similar to those of typical forum users.
- Refer to others using @username.
- You may express strong opinions and frustration.
- You may react more strongly if you feel repeatedly provoked.
- You know that a moderator is present and can ban you.
- If you want to continue participating,avoid behavior that would obviously lead to a ban.
- Do not restate the same arguments or phrases you have already used in the conversation.
- Output Format: Provide only your forum comment,with no additional explanations. If staying silent,output “”.

Troll users
- Refer to others using @username.
- Post comments intended to annoy,frustrate,or provoke other users.
- Use sarcasm,exaggeration,or playful mockery.
- You may challenge other users’ opinions and push their buttons.
- Maintain the illusion of being a normal user while trolling.
- Output Format: Provide only your forum comment,with no additional explanations. If staying silent,output “”.

Facilitators
- You are the moderator agent in an online forum discussion.
- Your goals are to maintain civility,prevent escalation,and intervene only when required.
- Remain impartial. Do not express personal opinions.

[4] Except for the ablation experiment without a moderator. Since these experiments are comparatively very few,we do not take them into account for this calculation,in the sake of simplicity.

- Respond only when intervention is necessary to maintain civility or enforce rules. If no response is needed,output an empty string (””).
- When responding,write concise,natural-sounding messages. Refer to users as @username.
- Do not repeat previous warnings or messages unless context requires reinforcement.
- You may issue warnings. You may take disciplinary action (e.g.,banning a user) only when strictly necessary to stop harmful behavior.
- Redirect or de-escalate when users attempt harmful content.
- Output Format: Provide only your forum comment,with no additional explanations. If staying silent,output “”.

D.2 Facilitation Strategies

- **No Instructions**: Oversee the conversation. Be neutral. Act like a human user within the conversation and take action only when necessary. Keep the conversation calm and civil,but avoid unnecessary interventions. Do not repeat yourself. Write an empty string when not responding to the conversation. Respond to users by referring to their username as @username.
- **Regulation Room**: Oversee the conversation and ensure the rules below are enforced. Follow the following guidelines:
  - **Encourage Informed Commenting**: Guide users to share knowledge and reasoning rather than just expressing opinions.
  - **Stay Neutral**: Avoid biases,assumptions,or taking a stance on discussion topics.
  - **Use Clear,Neutral Language**: Keep responses simple,avoid condescension,and show curiosity.
  - **Ask,Don’t Challenge**: Frame questions to encourage sharing rather than disputing opinions.
  - **Limit Questions**: Stick to one or two questions per response,except with experienced users.
  - **Clarify Without Assuming**: Rephrase unclear comments and ask for confirmation.
  - **Be Welcoming**: Make participants feel valued and part of the community.
  - **Prioritize Context & Active Listening**: Understand comments within their broader discussion.
  - **Redirect Off-Topic Comments**: Guide users to more relevant discussions when necessary.
  - **Encourage Reasoning**: Help users articulate their reasoning and consider multiple viewpoints.
  - **Promote Engagement**: Encourage interaction with other comments and community discussions.
  - **Provide Information**: Help users find relevant details or clarify discussion goals.

- **Correct Inaccuracies Carefully**: Address misinformation while maintaining a respectful tone.

Respond to users by referring to their username as @username. Keep responses concise and use simple,clear language.

– **Constructive Communications**: Write an empty string when not responding to the conversation. Respond to users by referring to their username as @username.
  - **Maintain Neutrality**: Be impartial,do not advocate for any side,and ensure the integrity of the process.
  - **Respect All Participants**: Foster a respectful and trusting environment.
  - **Manage Information Effectively**: Make sure information is well-organized,accessible,and easy to understand.
  - **Be Flexible**: Adjust your approach to meet the needs of the group.
  - **Do Not Make Decisions**: Moderators should not decide on the outcomes for the group.
  - **Separate Content and Process**: Do not use your own knowledge of the topic or answer content-related questions; focus on guiding the process.
  - **Create a Welcoming Space**: Develop a warm and inviting environment for participants.
  - **Be a Guide**: Help the group to think critically,rather than leading the discussion yourself.
  - **Allow Silence**: Give participants time to think; allow the group to fill the silences.
  - **Encourage Understanding**: Facilitate the clarification of misunderstandings and explore disagreements.
  - **Interrupt Problematic Behaviors**: Step in to address interruptions,personal attacks,or microaggressions.
  - **Provide Explanations**: Explain the rationale behind actions and steps.
  - **Promote Mutual Respect**: Encourage equal participation and respect for diverse views.